

Date	July 1, 2026
Team ID	SB-AI-2026-9284
Project Name	Sustainable Waste Management AI Assistant
Maximum Marks	2 Marks

Solution Requirements Specification

1. Functional Requirements

- FR-1: Search-based AI classification mapping text strings to strict JSON structures.
- FR-2: Dynamic categorization (Plastic, Paper, Glass, Organic, Metal, Hazardous, E-waste, Medical waste).
- FR-3: Detailed protocol parser showing hazard level, cleaning guide, and reusable recommendations.
- FR-4: Interactive Leaflet map loading geo-pins filtered by center type.
- FR-5: History stack with delete functions and dashboard analytics graphs (Line, Doughnut, Bar).

2. Non-Functional Requirements

- NFR-1: Load speed - less than 2 seconds.
- NFR-2: Resilient Local Fallback - offline database and heuristics scanner if keys are not set.
- NFR-3: Dark Mode design system using responsive glassmorphic CSS rules.